Throughput Accounting + DBR Scheduling + Order Acceptance Design (v3.28)

Nature of this document: Cross-domain methodology paper spanning **Operations Research / Managerial Accounting / ERP / AI**, describing erpilot v3.28's **Throughput Accounting (TA)** + **Drum-Buffer-Rope (DBR) Scheduling** + **Order Acceptance Decision** modules. This release completes Goldratt's (1984) TOC trilogy: v3.27 IDENTIFIED bottleneck → v3.28 EXPLOITS (DBR) + SUBORDINATES (TA).

📘 Prerequisites: `MRP_ALGORITHM_DESIGN_EN.md` (v3.25.10) / `MRP_CAPACITY_AWARE_DESIGN_EN.md` (v3.26) / `PLANNING_EXPLAINABILITY_DESIGN_EN.md` (v3.27)

Abstract

This paper describes erpilot v3.28's completion of Goldratt's (1984) Theory of Constraints trilogy's latter two steps: (i) **Throughput Accounting (TA)** [Goldratt 1992; Corbett 1998] — replacing traditional cost accounting for product-mix decisions; (ii) **Drum-Buffer-Rope (DBR) Scheduling** [Schragenheim & Dettmer 2000] — using bottleneck (CCR) as the production pacemaker; (iii) **Order Acceptance Decision** — implementing Goldratt 1990 §6's "highest T/CCR-min first" rule. We further prove this rule is **optimal under a single binding constraint** (continuous knapsack relaxation argument) and validate with 21 structural-invariant tests including 4 canonical Goldratt textbook cases. SMB owners' daily question "**Should we take this order?**" can be answered by `evaluate_order_acceptance()` in ~10 ms.

Keywords: Throughput Accounting, DBR, order acceptance, product mix, TOC, bottleneck pricing

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1. Introduction

1.1 Why Traditional Cost Accounting Misleads Decisions

Standard cost accounting allocates fixed overhead per unit (direct labor hours, machine depreciation), yielding a "unit cost". Under this framework, if an order's price < unit cost, it's judged "loss-making" and rejected.

However **Goldratt (1992)** *The Haystack Syndrome* points out the fundamental error:

> *"Fixed overhead won't actually cost more because of this order. If workers are already on payroll and machines already depreciating, this order using some of their time doesn't generate extra overhead."*

Concrete example: product standard cost $80 (material $30 + labor allocation $40 + overhead $10), selling price $75. Standard accounting: "loses $5", reject. TA analysis:

Item	Standard Cost	Throughput Accounting
Revenue	$75	$75
TVC (truly variable cost)	n/a	$30 (material only)
Labor + overhead allocation	$50	$0 (period fixed cost)
"Profit"	-$5 (reject)	$45 throughput (take it!)

The accept-vs-reject difference = $45 × orders × 12 months = **substantial real cash flow**. Standard cost accounting **misjudges** → customers wrongly reject high-throughput orders, accumulating large losses yearly.

1.2 TOC Trilogy Integration

Step	Source	erpilot Module
IDENTIFY constraint	Goldratt 1984	v3.27 `identify_bottlenecks`
EXPLOIT constraint	Goldratt 1990	v3.28 DBR + ranking
SUBORDINATE to exploit	Goldratt 1990	v3.28 TA decision
ELEVATE constraint	Goldratt 1990	v3.27 counterfactual + OT/alt
REPEAT	Goldratt 1990	Continuous loop

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2. Methodology

2.1 Module 1: Throughput Accounting Three Variables

Goldratt 1992 defines:

$$ \text{Throughput (T)} = \text{Revenue} - \text{Truly Variable Cost (TVC)} $$

TVC strict definition: cost must vary **1:1 with product quantity**. Includes:

Counts as TVC	Does NOT count (is OE)
✅ Raw materials (BOM components × scrap factor)	❌ Direct labor (workers' wages fixed)
✅ Sales commission (% of revenue)	❌ Machine depreciation
✅ Outsourcing fees	❌ Factory rent
✅ Packaging consumables	❌ Supervisor / management salaries

Operating Expense (OE) = period fixed costs.

Inventory (I) = WIP + raw + finished (at TVC, **excluding overhead allocation**).

Net Profit = Σ T − OE (no per-unit fixed overhead calculation needed).

2.2 Module 2: Order Acceptance Decision (The Killer Decision)

For a single order (p, q, P) (product, quantity, unit price):

$$ T_{\text{order}} = q \cdot (P - \text{TVC}_p(P)) $$

$$ \text{T per CCR min} = \frac{T_{\text{order}}}{q \cdot \text{bottleneck\_min}_p} $$

Goldratt 1990 §6 decision rule:

```
If T_order < 0:        REJECT     (loss-making)
If bottleneck full:    REJECT     (cannot deliver)
If T/CCR-min < threshold:  NEGOTIATE  (raise price or outsource)
Otherwise:             ACCEPT
```

Continuous Knapsack Optimality Argument: under single binding constraint $\sum_i q_i \cdot a_i \leq C$, maximizing $\sum_i q_i \cdot t_i$ has LP relaxation optimum given by "sort by t_i/a_i descending, fill capacity greedily". **Goldratt 1990 §6** proves this rule is optimal for single CCR. Multi-CCR degenerates to LP (future sprint with PuLP).

2.3 Module 3: Drum-Buffer-Rope Scheduling

Schragenheim & Dettmer (2000) *Manufacturing at Warp Speed* three elements:

Element	Meaning	Calculation
🥁 Drum	Bottleneck pacemaker	$\text{rate} = C_{\text{CCR}} / \text{run_time}_{\text{CCR}}$
🛡 Buffer	Time buffer before bottleneck (anti-starvation)	$\text{buffer} = 3 \times \text{run_time}_{\text{CCR}}$
🧬 Rope	Material release sync	$\text{release_offset} = \text{buffer_time}$

Why buffer = 3× run-time: Schragenheim 2000 §4 empirically observed 3× balances "starvation risk" vs "WIP cost" in typical SMB environments. Theoretically:

- $1\times$: too small, slight upstream delay → bottleneck starves
- $3\times$: typical optimum, acceptable WIP cost
- $5\times+$: over-protection, WIP bloat

Hopp & Spearman 1996 *Factory Physics* §10: throughput is determined by bottleneck only; buffers at non-bottleneck stations are pure waste. erpilot's DBR module places buffer only before bottleneck.

2.4 Module 4: Pricing Curve (Sensitivity Extension)

For given (product, qty, base_price), scan discount levels (e.g., [0, 5%, 10%, 15%, 20%]), compute throughput, T/CCR-min, recommendation change per scenario.

Sales use case: when customer negotiates, sales can answer instantly "5% off OK", "10% needs supervisor", "15% reject".

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3. Implementation

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backend/app/services/throughput_accounting.py  (~450 lines)
├── TVCBreakdown (dataclass)
├── compute_product_tvc(db, product_id, commission, outsourcing)
├── OrderEvaluation (dataclass with throughput / t_per_ccr_min / recommendation)
├── compute_throughput_per_ccr(rev, tvc, bn_min) → float       ← Goldratt killer metric
├── evaluate_order_acceptance(...) → OrderEvaluation            ← main decision
├── PricingScenario / explore_pricing_curve(...)                ← what-if discounts
├── DBRSchedule / compute_dbr_schedule(...)                     ← Schragenheim DBR
├── rank_orders_by_t_per_ccr(orders)                            ← Goldratt §6 ordering
└── select_best_product_mix(orders, total_capacity)             ← LP-optimal greedy
```

See `THROUGHPUT_ACCOUNTING_DESIGN_ZH.md` §3 for the architecture integration diagram.

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4. Validation

4.1 Seven Categories of Structural Invariants (21 tests, all pass)

1. **TVC composition** — Goldratt 1992 strict definition (3 tests)
2. **T/CCR-min monotonicity** — basic / zero / monotone in price (3 tests)
3. **Order decision logic** — reject infeasible / reject loss / accept / negotiate / no-CCR (5 tests)
4. **DBR Schragenheim 2000 rules** — 3× buffer / drum rate / degenerate / custom (4 tests)
5. **Knapsack optimality** — Goldratt §6 ranking / capacity respect / infeasible ordering (3 tests)
6. **Pricing curve** — monotonicity / recommendation flip (2 tests)
7. **DB integration** — BOM-derived TVC = 26.0 known answer (1 test)

4.2 Results

21/21 tests pass. Sprint cumulative: **412/412 smoke tests pass**.

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5. Limitations and Future Work

Topic	Why not	Future direction
Multi-CCR binding	Greedy no longer optimal; need LP/MIP	v3.29: integrate PuLP+CBC
OE auto-derivation	Currently customer-input	Integrate HR + Fixed Asset modules
Dynamic buffer management	Fixed 3× currently	Schragenheim 2000 §6 buffer management
Stochastic lead-time in DBR	Deterministic assumption	Hopp-Spearman §6 stochastic LT
Multi-bottleneck schedule sync	Single CCR assumption	DBR-B-D sequence
Outsourcing detection in TVC	Manual entry	Routing.is_outsourced flag
GAAP/IFRS reconciliation	TA ≠ standard cost accounting	Cost-accounting bridge layer

Edge Cases

1. **Product with no BOM**: material_cost = 0; customer must manually set outsourcing_cost or evaluation unreliable
2. **Product with no Routing**: bottleneck_minutes_required = 0 → always accept (from CCR perspective); may be limited by other constraints (v3.29 multi-constraint)
3. **commission_rate > 1**: physically impossible, not checked (trust input)
4. **Negative price**: not checked (trust input) — should not occur in practice

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6. Legal Notice / 法律聲明

⚠️ Important: Legal Nature of Managerial Accounting Analysis

This module produces **managerial accounting analysis** per **Goldratt (1992) Throughput Accounting** framework. It is **NOT a GAAP-compliant or IFRS-compliant financial statement**.

1. TA vs Standard Cost Accounting Distinction

Use case	TA	Standard cost accounting
External financial reports / tax filings	❌ NOT applicable	✅ Required
Internal product-mix decisions	✅ Applicable	❌ Often misleads
Order acceptance decisions	✅ Applicable	❌ Often wrongly rejects high-T orders

Customers **must NOT** use this module's throughput / TVC output directly as basis for financial reports or tax filings. External reports must follow applicable accounting standards + CPA review.

2. Order Acceptance Responsibility

This module's `recommendation` (accept/reject/negotiate) is **an algorithm-computed suggestion based on input parameters**, NOT:

- A **commitment** to quote/take an order
- A **mandatory execution command** for sales
- A **legally binding** price

Before acting on recommendations, qualified business/finance/legal personnel must **manually review**:

- Whether **TVC classification** matches your company's accounting policy (varies by industry)
- Whether **commission_rate** matches sales contracts
- Whether **outsourcing_cost** includes outsourcing-contract quality/delivery penalties
- Whether **bottleneck_minutes_required** reflects **actual** situation (including setup, idle waiting, QC)
- Whether **min_acceptable_t_per_min** threshold accounts for **opportunity cost** (future demand)

3. DBR Scheduling Responsibility

Buffer = 3× run_time is Schragenheim 2000's **empirical value**, **NOT guaranteed** optimal in all environments:

- High upstream lead-time variability environments → 3× may be insufficient (increase)
- Low frequency, high product variety (high-mix-low-volume) environments → 3× may be excessive
- Customers should observe actual starvation / WIP and **self-adjust** `buffer_multiplier`

4. Pricing Curve Is Not Antitrust Commitment

Pricing scenarios **do NOT constitute**:

- Legal advice on **discriminatory pricing strategy** toward specific customers
- **Antitrust compliance** confirmation for market behavior

National antitrust laws (e.g., Taiwan Fair Trade Act, U.S. Sherman Act) regulate pricing behavior. **Different prices for different customers** may raise compliance issues. Consult legal counsel.

5. Algorithm Limitations

- **Single CCR assumption**: greedy algorithm is **not optimal** when multiple CCRs bind simultaneously; multi-bottleneck customers should use LP/MIP
- **Deterministic assumption**: bottleneck_minutes_required, TVC assumed deterministic; lead-time variability, raw-material price volatility not modeled
- **Continuous relaxation**: knapsack assumes order qty divisible; in practice 0-1 knapsack is NP-hard
- **No strategic interaction**: customer bargaining, competitor pricing response not modeled (game theory scope)

6. Disclaimer Clause

To the maximum extent permitted by applicable law, erpilot assumes no liability for:

- **Revenue losses or customer-relationship damage** from wrong accept/reject based on this module's recommendation

- **Financial misstatement** consequences from TVC/OE misclassification
- **Production delay / inventory excess** from improper DBR buffer settings
- **Antitrust disputes** from pricing-curve application
- **Contractual / labor / competition law** disputes by third parties (customers, suppliers, sales, operators) acting on these recommendations

7. Cumulative Applicability of Predecessors

This version overlays v3.25.10 + v3.26 + v3.27; **all disclaimers** in prerequisite documents §6 **apply cumulatively** here.

Recommended Practice

- Treat module output as "**business decision-support report**" — NOT "automatic accept/reject system"
- Major orders (e.g., > 5% of annual revenue) must go through management review
- Compare TVC settings vs actual costs quarterly; adjust parameters
- Partner with CPA to build TA → standard-cost reconciliation bridge
- Pricing-curve results should be considered alongside market conditions and competitive landscape

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8. Changelog

Version	Date	Change
v3.25.10	2026-05-20	Uncapacitated MRP-II
v3.26	2026-05-20	Capacity-aware MRP
v3.27	2026-05-20	TOC IDENTIFY + Provenance
v3.28	**2026-05-20**	**This release**: TOC EXPLOIT + SUBORDINATE (TA + DBR + Order Acceptance)
Future v3.29+	TBD	Multi-CCR LP/MIP / Dynamic buffer management
Future v3.30+	TBD	LLM wrapper "should we take this order" Q&A

Last updated: 2026-05-20 (v3.28) **Authors**: erpilot engineering team (with IE/OR/Managerial Accounting/AI cross-domain academic methodology) **Version**: 1.0 **Prerequisites**: v3.25.10 / v3.26 / v3.27 design docs **Chinese version**: `THROUGHPUT_ACCOUNTING_DESIGN_ZH.md`